

## **Graduation Thesis**

# **TRUSTAY-AI: A BOARDING HOUSE MANAGEMENT SYSTEM LEVERAGING MULTI-AGENT ARCHITECTURE AND RAG WITH NATURAL LANGUAGE KNOWLEDGE HARVESTING**

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**Major: Software Engineering**

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# PRESENTATION OUTLINE

**01** Overview  
(Problem Context & Solution)

**02** Techonology & Architecture  
(Tech Stack & System Model)

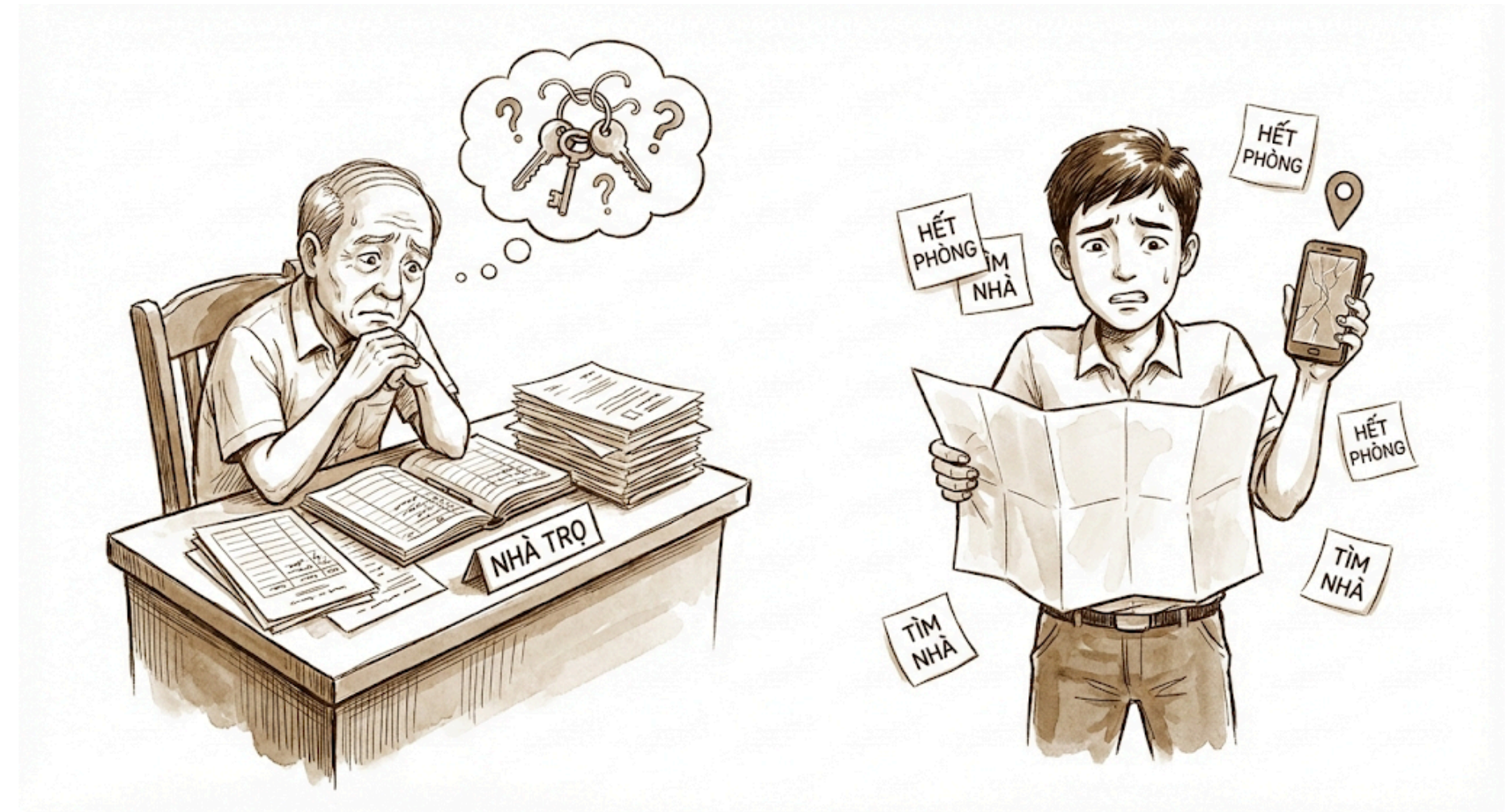
**03** System analysis & Design  
(Use Cases & Database Schem)

**04** Implementation  
(Key Features & Interface Demo)

**05** Result & Future work  
(Testing Outcomes & Next Steps)

## Current Problems:

- **Fragmented Management:** Data scattered across notebooks & Excel.
- **Manual Processes:** Paper contracts, cash payments, hard to reconcile.
- **Difficult Search:** Unreliable info, lack of intelligent filters.



## CONSEQUENCES

These inadequacies lead to inefficient operations, wasted time and costs, and a poor experience for both landlords and tenants.



## Trustay Solutions:

- **All-in-One Platform:** Centralized management for Landlords/Tenants.
- **Digitization:** Electronic contracts & automated payments.
- **AI-Integration:** Natural language search, help Landlords create room faster.
- **Platform:** Website & Zalo Mini App





### Backend & Core

#### NestJS + Prisma + PostgreSQL

- Enterprise-grade modular architecture, highly scalable.
- Type-safe ORM ensures fully secure and reliable database queries.



### Frontend & Mobile

#### Next.js + Zalo Mini App

- Optimized SEO and fast page loading through SSR.
- Instantly reaches 75 million Zalo users without requiring installation.



### AI & Search (Key Tech)

#### RAG + Multi-Agent + Text2SQL + Elasticsearch

- Mitigates AI hallucinations and improves response accuracy.
- Supports fuzzy search with tolerance for spelling errors.



## Text2SQL

Text-to-SQL translates natural language into SQL, allowing you to directly 'talk' to your database without writing any code.

Who paid late in the last  
3 months?

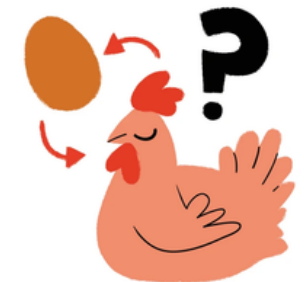
Sure! That would be Room 104, Room  
207, and Room 310.

TALK TO YOUR DATA

## 02. Technology & Architecture: Text2SQL

### Barrier to Text2SQL: Data Cost

- High-accuracy Text-to-SQL models require massive amounts of **"Golden Data"** (Human-annotated SQL pairs) for fine-tuning.
- Creating this dataset is expensive and time-consuming

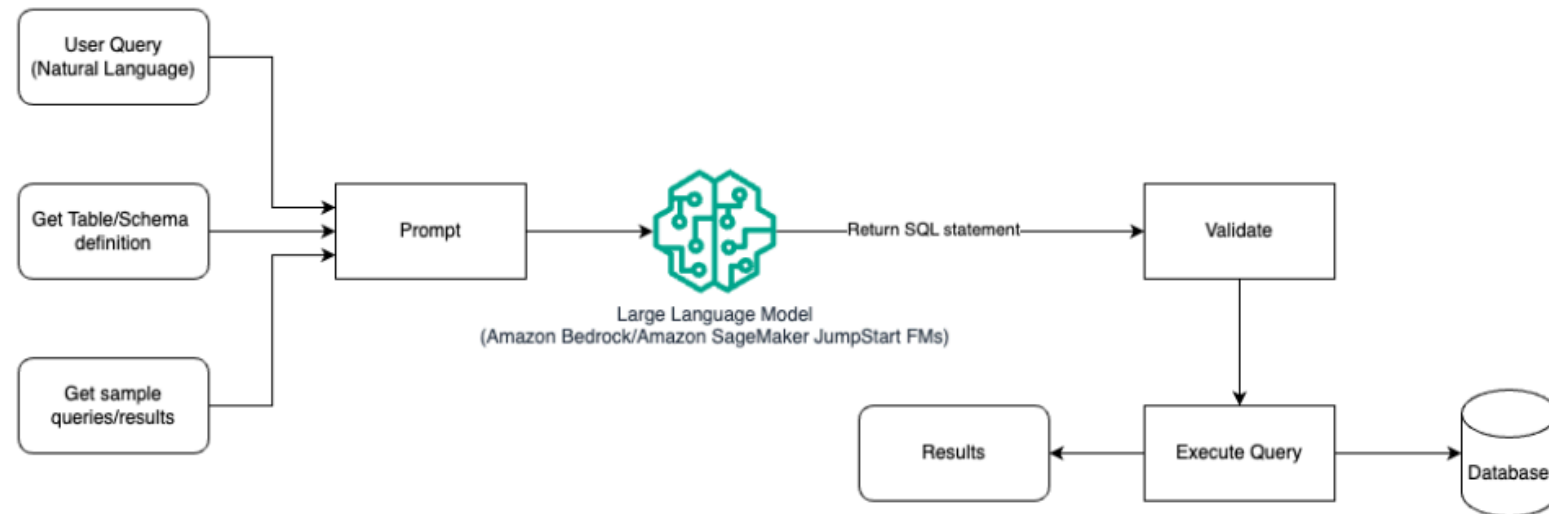


### The Cold Start Paradox

#### "No Data ↔ No Operation"

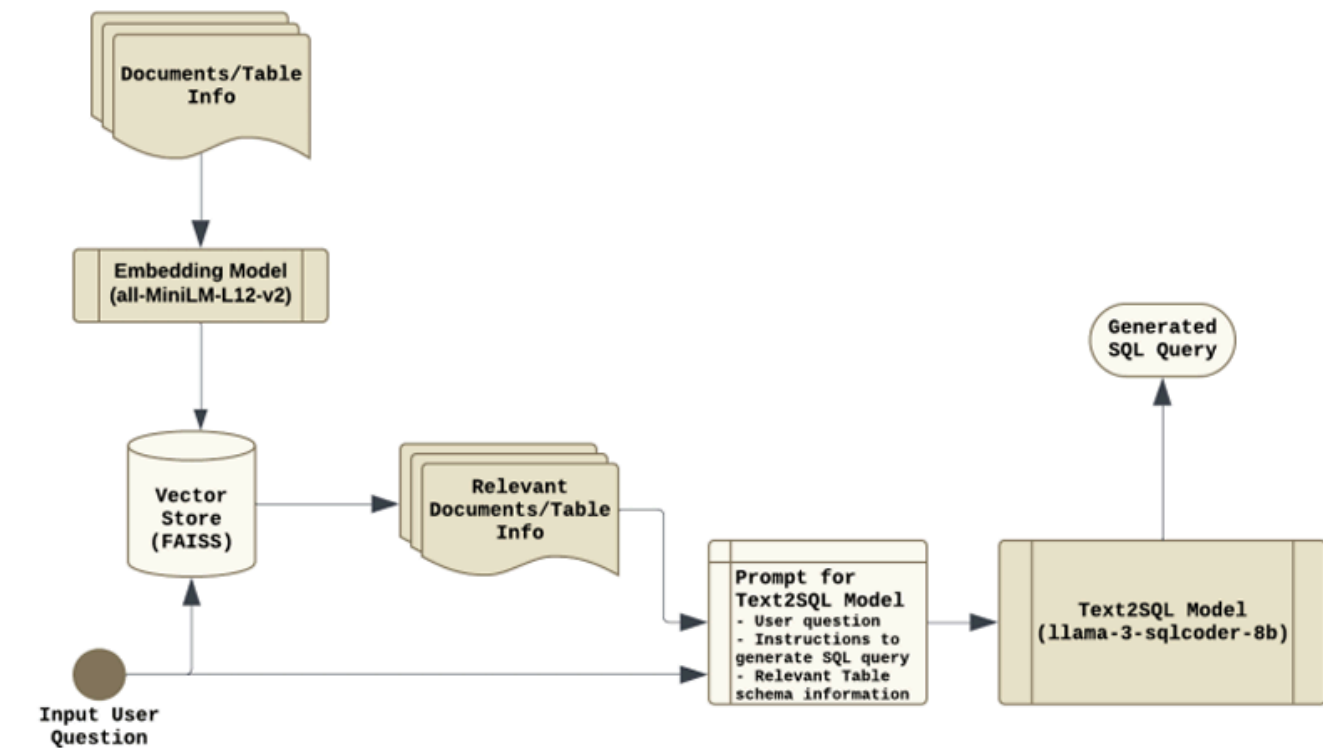
The system needs data to operate effectively, but it must operate to generate data.

### Text2SQL Prompt engineering



- ✓ No need golden data
- ✗ Context Overflow
- ✗ Security Vulnerabilities

### Text2SQL with RAG



- ✓ No need golden data
- ✓ Context Overflow
- ✓ Security Vulnerabilities
- ✗ Redundant Processing
- ✗ Knowledge Retention



### Trustay AI



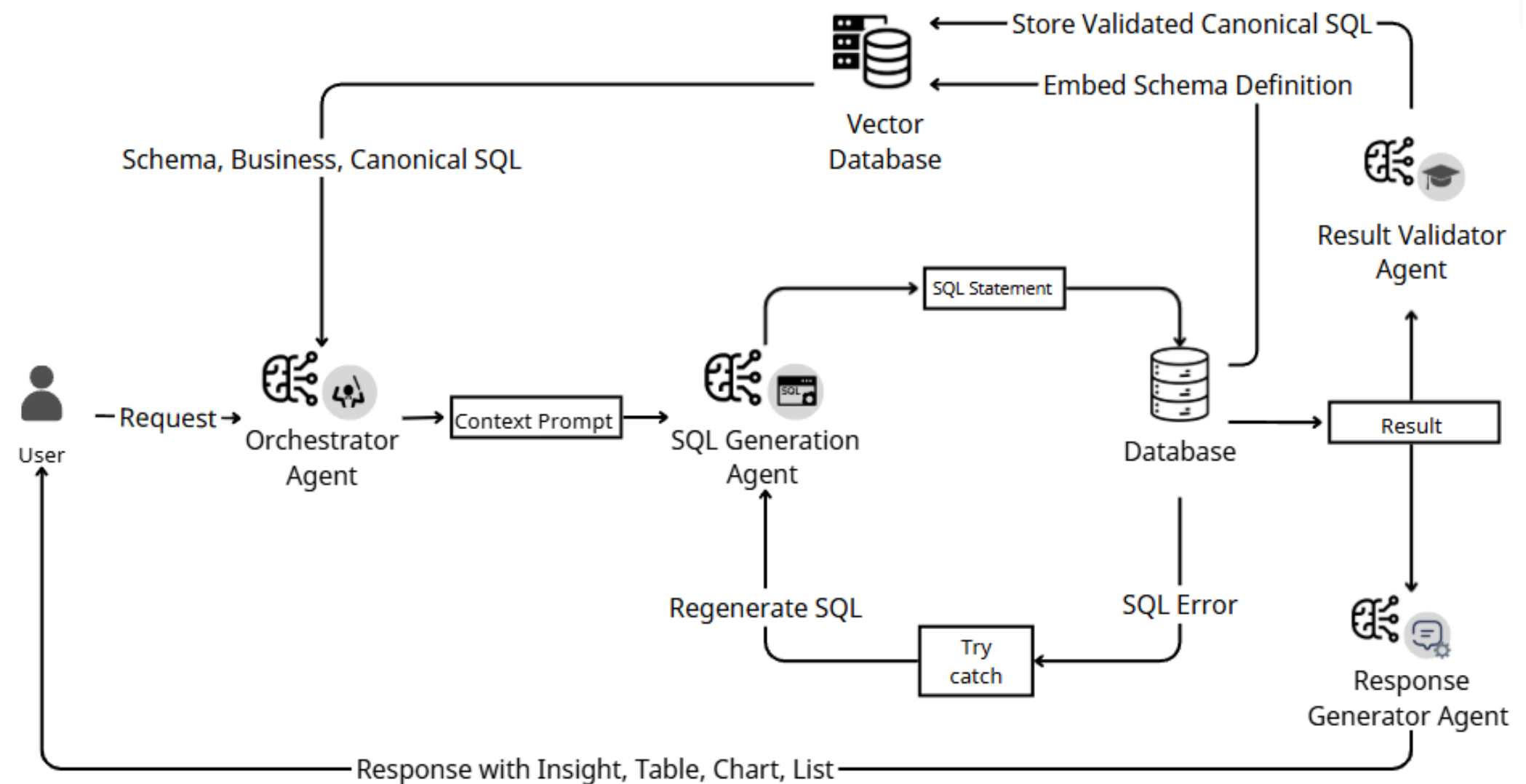
#### Solved 5 Listed Problems

**Orchestrator:** Understands user intent, fetch schema, canonical SQL info via **RAG**.

**SQL Generation:** Generates SQL code & performs Self-Correction..

**Response Agent:** Formats data into user-friendly responses.

**Validator Agent:** "Gatekeeper" that validates logic & semantics..



### Validate Knowledge

### Teaching Knowledge

### Logging System

ID: 0d696f06-5a6a-4167-849e-97ca13a26aac

Status: pending

Created: 17:21:13 17/12/2025

Updated: 17:21:13 17/12/2025

Session ID: 87f1877f-712b-48c0-bebc-7d2f87397e40

User ID: e36a0f86-b615-4f52-bd44-603fa7e92be7

Xem Log

Câu hỏi

Tìm phòng dưới 4 triệu rưỡi có chỗ đậu xe ô tô

SQL

```
SELECT r.id, r.name AS title, rp.base_price_monthly, 'room' AS entity
FROM rooms r
JOIN room_pricing rp ON rp.room_id = r.id
WHERE rp.base_price_monthly <= 4500000
AND EXISTS (
  SELECT 1
  FROM room_costs rc
  JOIN cost_type_templates ctt ON rc.cost_type_template_id = ctt.id
  WHERE rc.room_id = r.id
  AND ctt.name = 'Gửi xe ô tô'
)
LIMIT 100;
```

Response

```
{"message": "Rất tiếc là hiện tại mình chưa tìm thấy phòng nào đáp ứng cả hai tiêu chí giá dưới 4 triệu rưỡi và có chỗ đậu xe ô tô. Bạn có muốn mình mở rộng phạm vi tìm kiếm một chút không? Ví dụ như tăng mức giá lên một chút hoặc tìm ở khu vực lân cận chẳng hạn?","payload":
```

Evaluation

SQL truy vấn đúng bảng `rooms` và `room\_pricing` để tìm phòng có giá dưới 4.5 triệu. Điều kiện tìm kiếm chỗ đậu xe ô tô được thực hiện bằng cách kiểm tra sự tồn tại của chi phí "Gửi xe ô tô" trong bảng `room\_costs`. Kết quả trả về 0 phòng, điều này có thể do không có phòng nào thỏa mãn cả hai điều kiện. Tuy nhiên, SQL vẫn đúng và hợp lý.

Validator Data

Valid: true

Reason:  
OK  
EVALUATION: SQL truy vấn đúng bảng `rooms` và `room\_pricing` để tìm phòng có giá dưới 4.5 triệu. Điều kiện tìm kiếm chỗ đậu xe ô tô được thực hiện bằng cách kiểm tra sự tồn tại của chi phí "Gửi xe ô tô" trong bảng `room\_costs`. Kết quả trả về 0 phòng, điều này có thể do không có phòng nào thỏa mãn cả hai điều kiện. Tuy nhiên, SQL vẫn đúng và hợp lý.

Violations:  
• []

Token Usage:  
Prompt: 0

Đóng

Accept

Reject

### Trustay AI

That how we solve the **Cold Start** problem

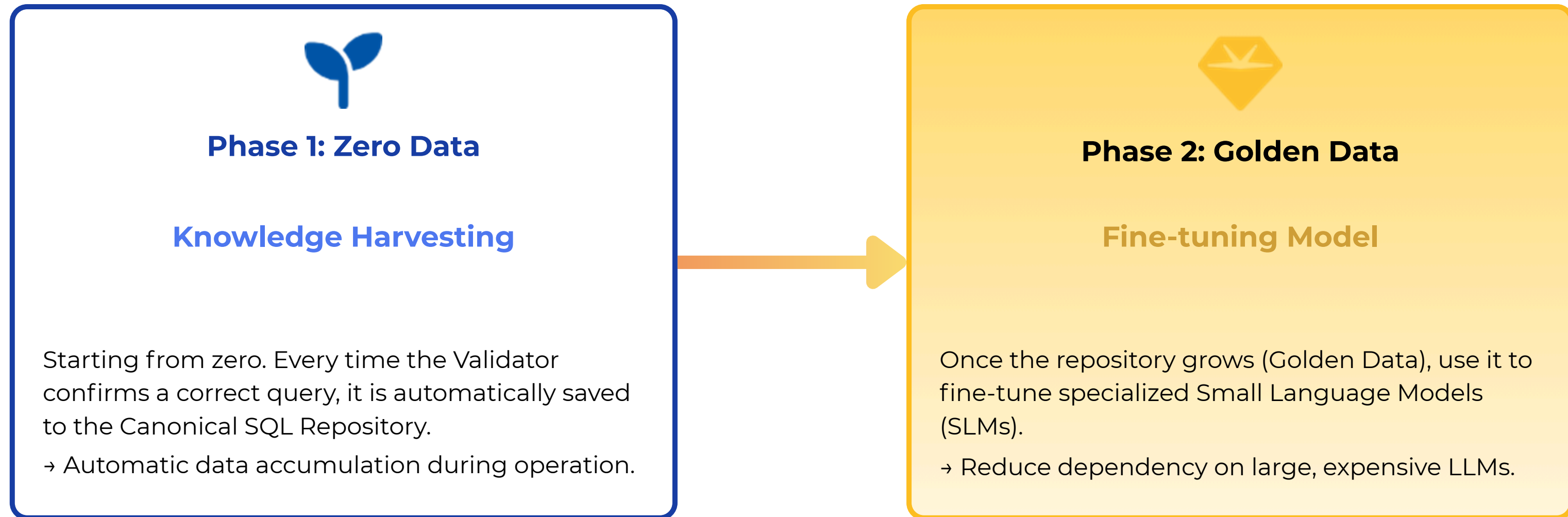
### The Cold Start Paradox After Trustay AI

~~"No Data ↔ No Operation"~~



No Data ↔ Can Operation → Collect Data

|     |                                                                                |                                                                                                                                                                                                                                                                                  |
|-----|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 291 | Tổng số tiền tôi đã thanh toán cho chủ nhà từ đầu năm đến giờ<br>Params: {}    | <pre>SELECT COALESCE(SUM(p.amount),0) AS total_paid_ytd FROM payments p JOIN rentals rnt ON rnt.id =</pre> <p>Template: select coalesce(sum(p.amount),0) as total_paid_ytd from payments p join rentals rnt on rnt.id = p.rental_id where rnt.tenant_id = :ten...</p>            |
| 290 | Giá điện hiện tại chủ nhà đang tính là bao nhiêu một số?<br>Params: {}         | <pre>SELECT rc.unit_price, rc.unit FROM rentals rnt JOIN room_instances ri ON ri.id = rnt.room_instar</pre> <p>Template: select rc.unit_price, rc.unit from rentals rnt join room_instances ri on ri.id = rnt.room_instance_id join rooms rm on rm.id = ri.room_id...</p>        |
| 289 | Tháng này tôi dùng hết bao nhiêu số điện?<br>Params: {}                        | <pre>SELECT SUM(COALESCE(rimr.meter_reading,0) - COALESCE(rimr.last_meter_reading,0)) AS kwh_used FRO</pre> <p>Template: select sum(coalesce(rimr.meter_reading,0) - coalesce(rimr.last_meter_reading,0)) as kwh_used from room_instance_meter_readings ri...</p>                |
| 288 | Kiểm tra xem khoản chuyển khoản hôm qua của tôi đã được xác nh:<br>Params: {}  | <pre>SELECT p.id, p.amount, p.payment_status, p.payment_date FROM payments p JOIN rentals rnt ON rnt.</pre> <p>Template: select p.id, p.amount, p.payment_status, p.payment_date from payments p join rentals rnt on rnt.id = p.rental_id where rnt.tenant_id ...</p>            |
| 287 | Hạn chót thanh toán của hóa đơn tháng này là ngày mấy?<br>Params: {}           | <pre>SELECT bll.due_date FROM bills bll JOIN rentals rnt ON rnt.id = bll.rental_id WHERE rnt.tenant_id</pre> <p>Template: select bll.due_date from bills bll join rentals rnt on rnt.id = bll.rental_id where rnt.tenant_id = :tenant_id and rnt.status = 'active' and ...</p>   |
| 286 | Tôi còn nợ tiền phòng tháng nào chưa đóng không?<br>Params: {}                 | <pre>SELECT bll.billing_period, bll.total_amount, bll.paid_amount, bll.remaining_amount, bll.status F</pre> <p>Template: select bll.billing_period, bll.total_amount, bll.paid_amount, bll.remaining_amount, bll.status from bills bll join rentals rnt on rnt.id = bl...</p>    |
| 285 | Cho tôi xem chi tiết tiền điện và tiền nước tháng 11<br>Params: {}             | <pre>SELECT bi.item_name, bi.amount, bll.billing_month, bll.billing_year FROM bill_items bi JOIN bill</pre> <p>Template: select bi.item_name, bi.amount, bll.billing_month, bll.billing_year from bill_items bi join bills bll on bll.id = bi.bill_id join rentals rnt on...</p> |
| 284 | Hóa đơn tiền nhà tháng này của tôi là bao nhiêu?<br>Params: {}                 | <pre>SELECT bll.id, bll.total_amount, bll.paid_amount, bll.remaining_amount FROM bills bll JOIN rente</pre> <p>Template: select bll.id, bll.total_amount, bll.paid_amount, bll.remaining_amount from bills bll join rentals rnt on rnt.id = bll.rental_id where rnt...</p>       |
| 283 | Liệt kê các bài đăng tìm bạn cùng phòng vừa mới đăng hôm qua<br>Params: {}     | <pre>SELECT rsp.id, rsp.title, rsp.monthly_rent, rsp.created_at FROM roommate_seeking_posts rsp WHERE</pre> <p>Template: select rsp.id, rsp.title, rsp.monthly_rent, rsp.created_at from roommate_seeking_posts rsp where rsp.status = 'active' and rsp.is_active...</p>         |
| 282 | Tìm các phòng đang thiếu 1 slot ở ghép dạng Ký túc xá<br>Params: {}            | <pre>SELECT rsp.id, rsp.title, rsp.remaining_slots, ri.room_number FROM roommate_seeking_posts rsp JO</pre> <p>Template: select rsp.id, rsp.title, rsp.remaining_slots, ri.room_number from roommate_seeking_posts rsp join room_instances ri on ri.id = rsp.ro...</p>           |
| 281 | Có bài đăng tìm roommate nào yêu cầu người đi làm công sở không?<br>Params: {} | <pre>SELECT rsp.id, rsp.title, rsp.additional_requirements, rsp.monthly_rent FROM roommate_seeking_pc</pre> <p>Template: select rsp.id, rsp.title, rsp.additional_requirements, rsp.monthly_rent from roommate_seeking_posts rsp where rsp.status = 'active' an...</p>           |





## 02. Technology & Architecture: Zalo Mini App

### Reasons for Choosing Zalo Mini App:

- **Market Reach:** Instantly access 75 million Zalo users.
- **User Psychology:** Eliminates the barrier of “reluctance to install apps,” especially for short-term tenants.
- **Cost Efficiency:** Saves development time and budget compared to building native Android/iOS applications.

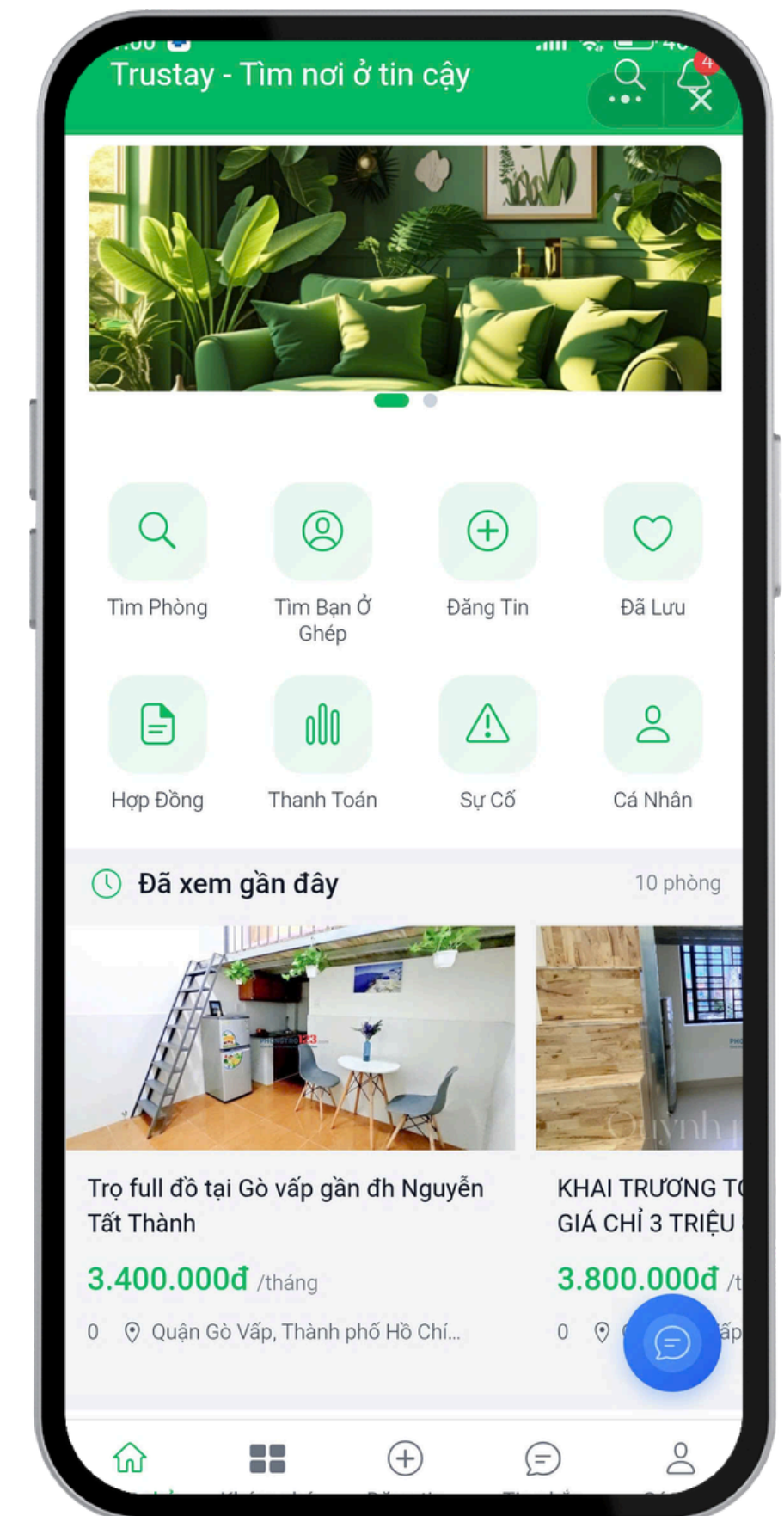
### Advantages:

- Instant access via QR code scanning — no installation required.
- Leverages the Zalo ecosystem, including ZaloPay payments and ZNS notifications.
- Ensures user credibility through verified, real-identity Zalo accounts.

Scan to use



Quét QR để truy cập Mini App



### Architecture Diagram

**Client Layer (Frontend):** Instant user access via Zalo Mini App and Next.js Web

### Processing Layer (Backend):

**NestJS + Prisma:** Enterprise-grade modular architecture ensuring data security.

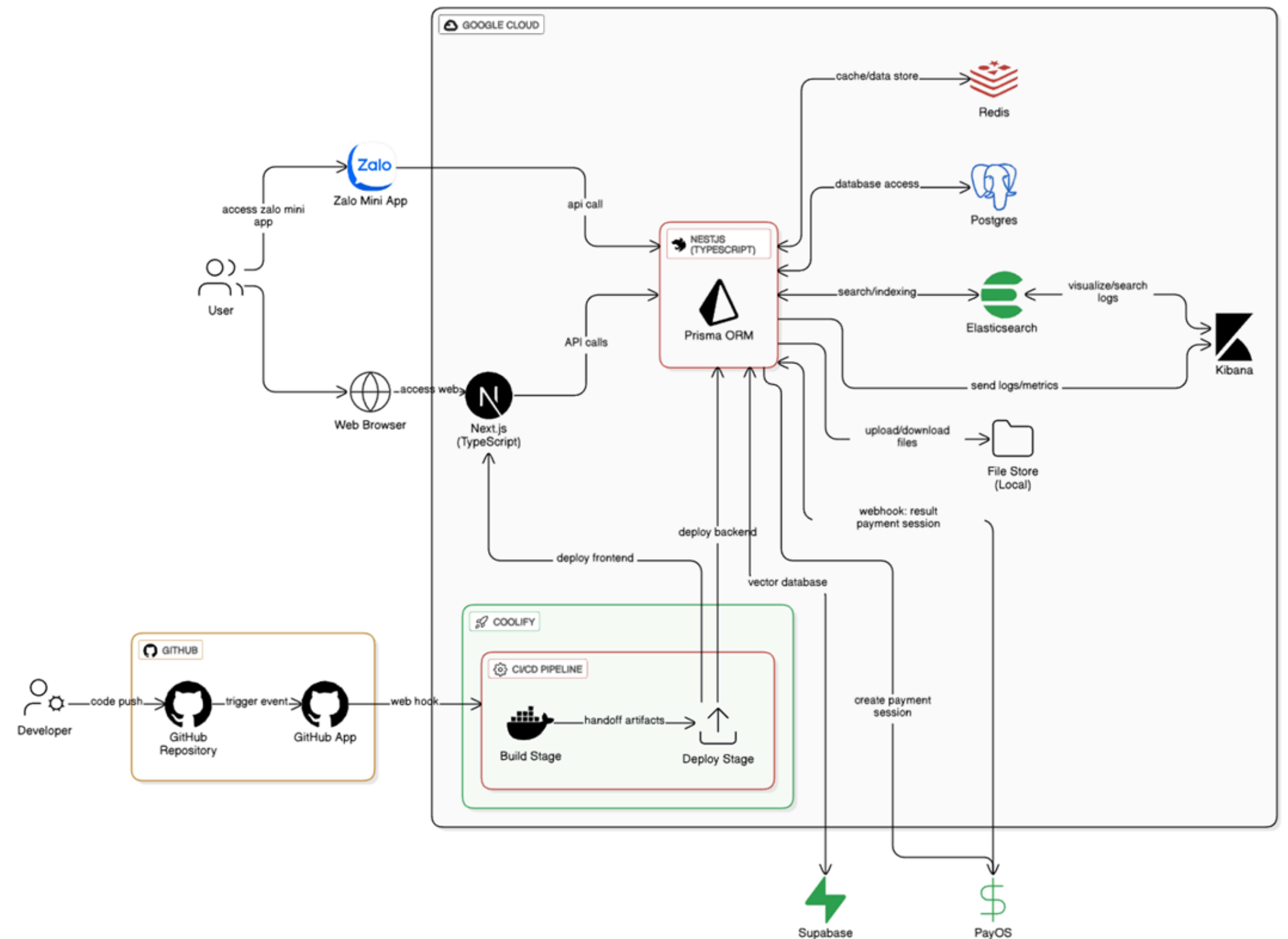
**Multi-Agent AI Flow:** for handling complex semantic queries

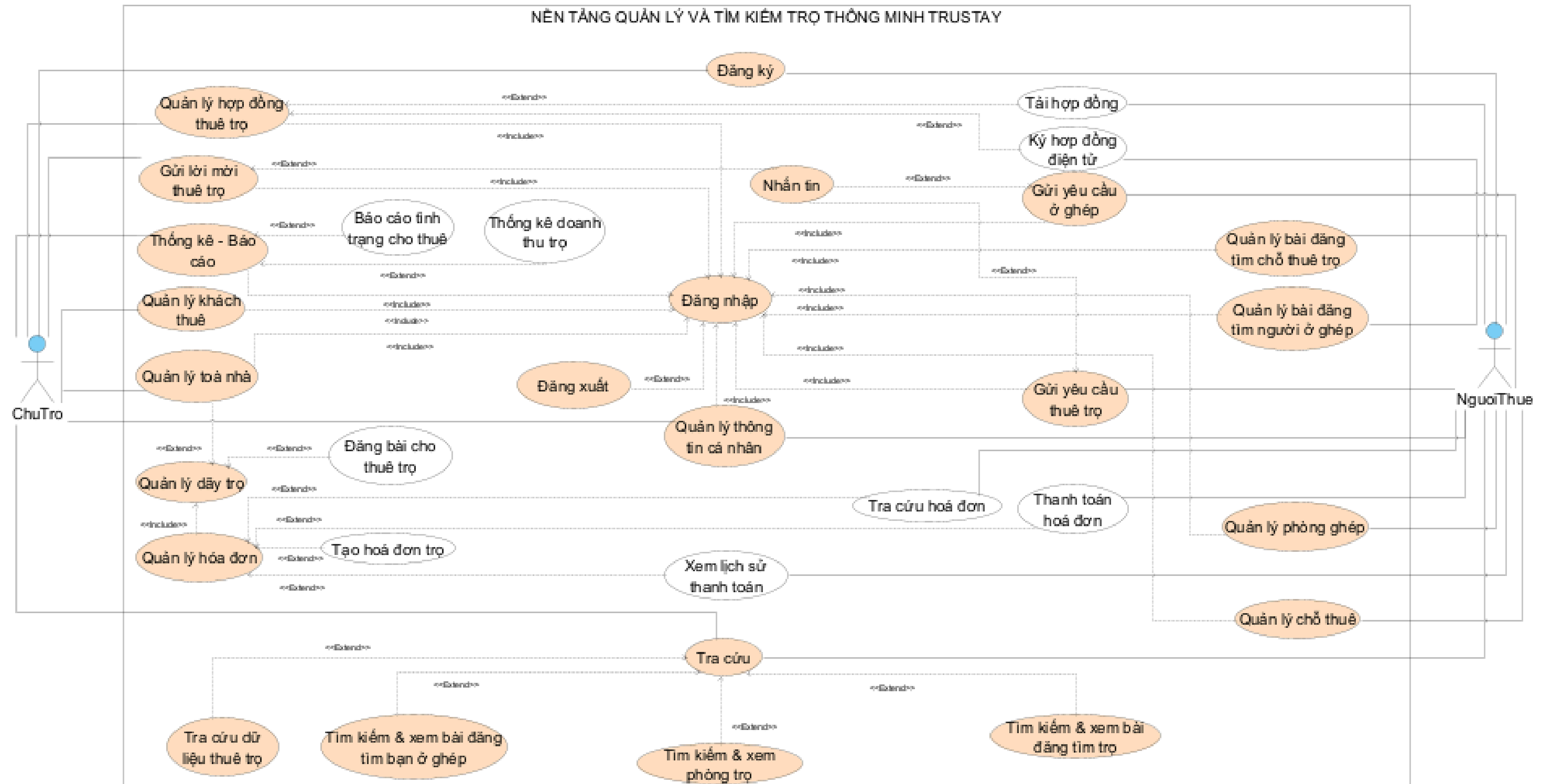
### Data & Performance Layer:

**Elasticsearch:** High-speed fuzzy search (< 100ms)

**Redis Cache:** Offloads the database and accelerates API responses

**Infrastructure (DevOps):** Deployed using Docker & Coolify, enabling fully automated CI/CD



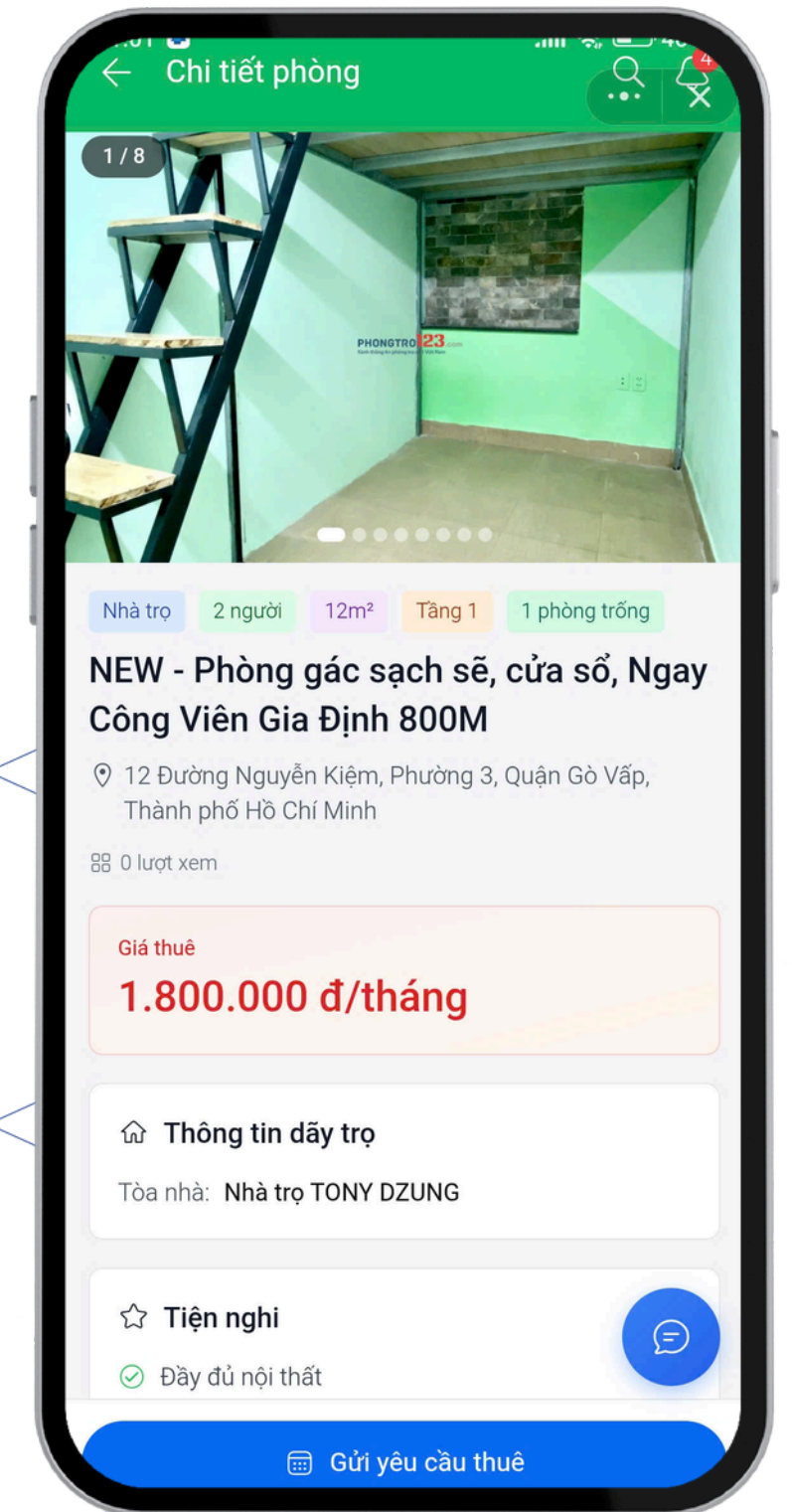
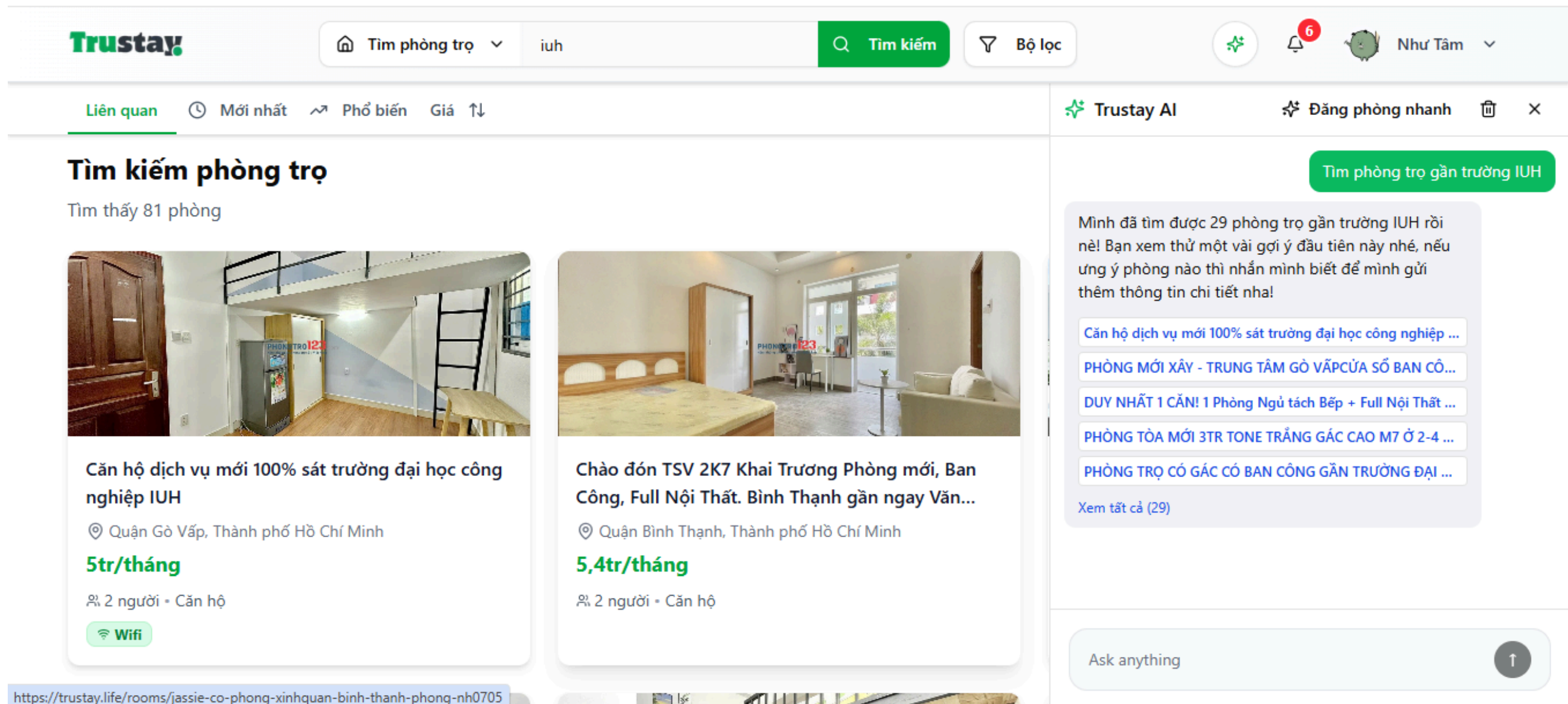




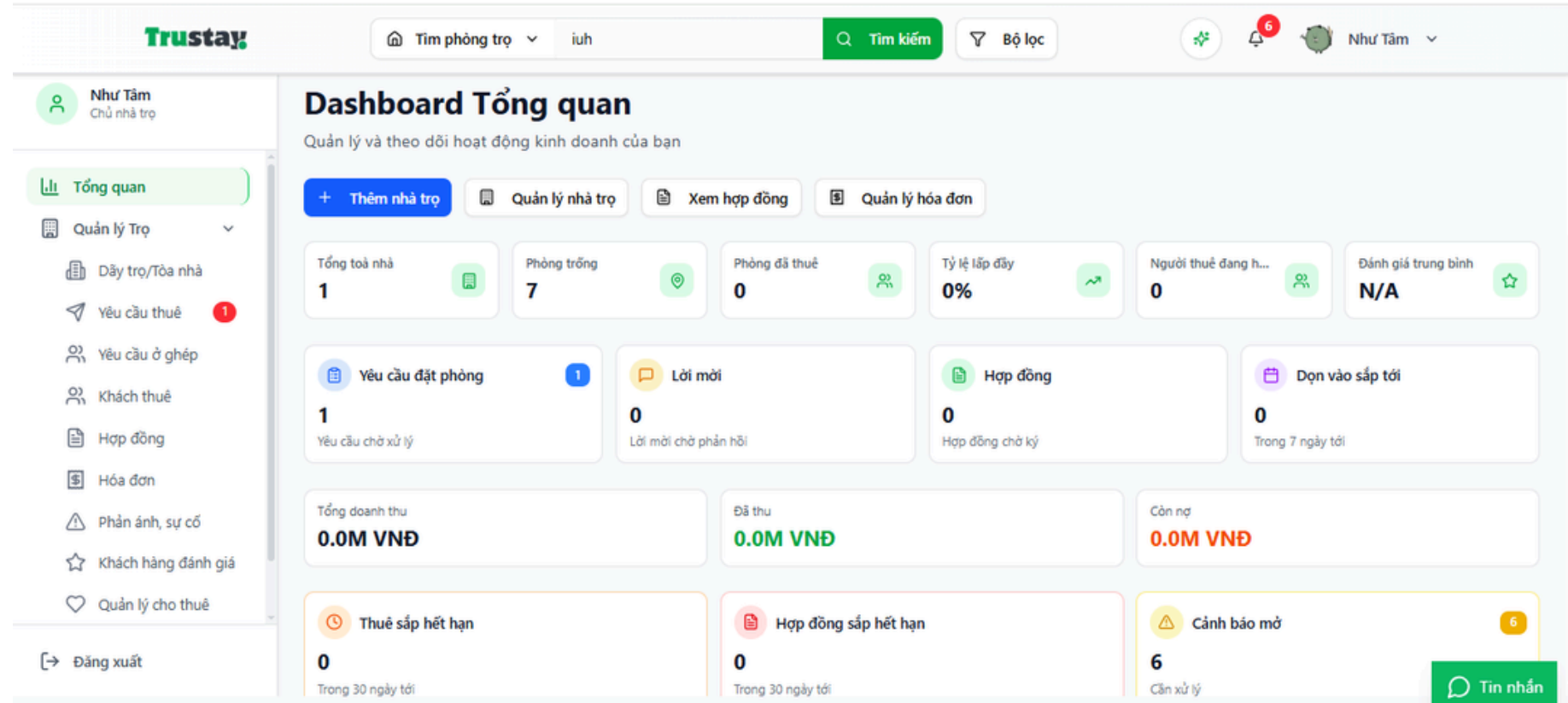




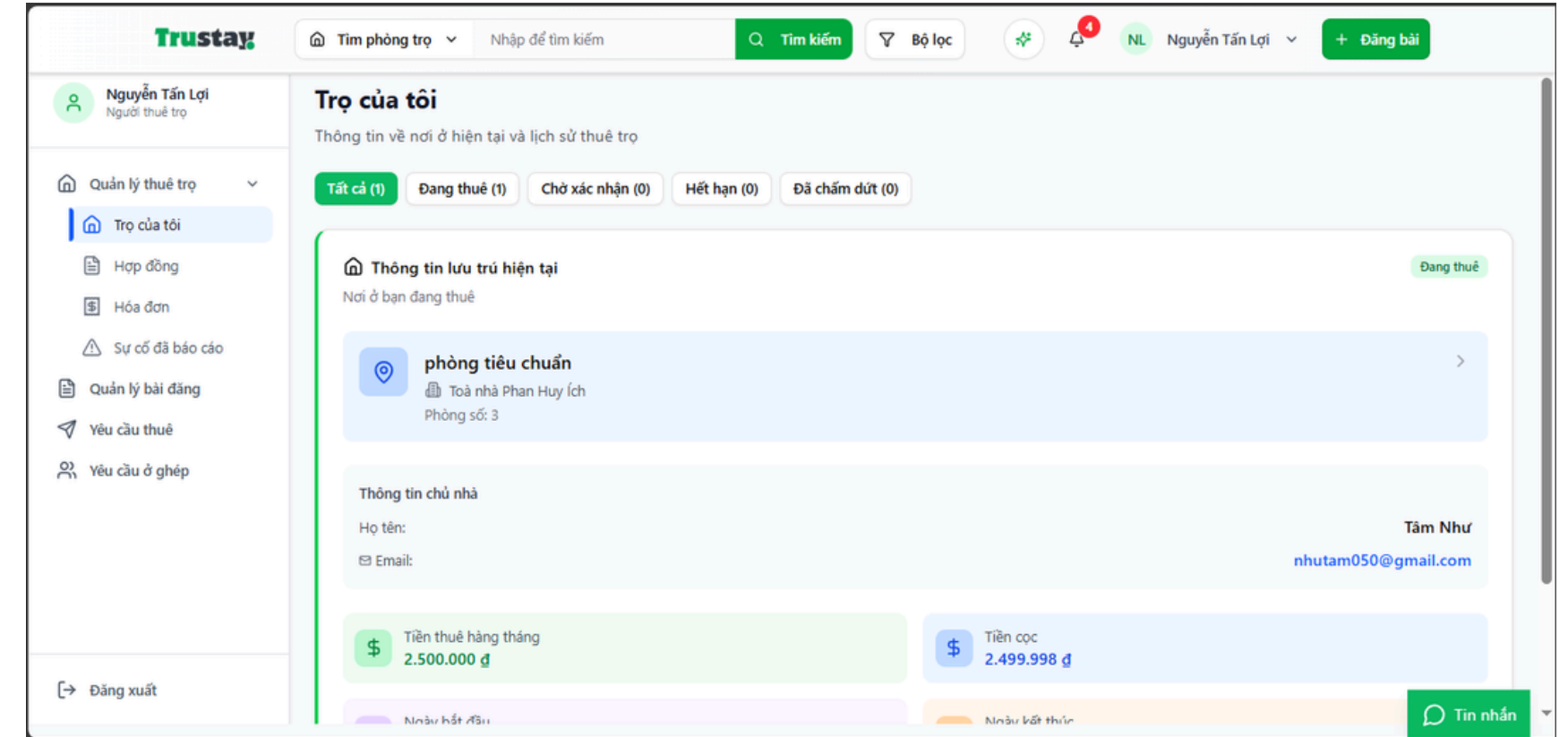




Search and Room Detail screen

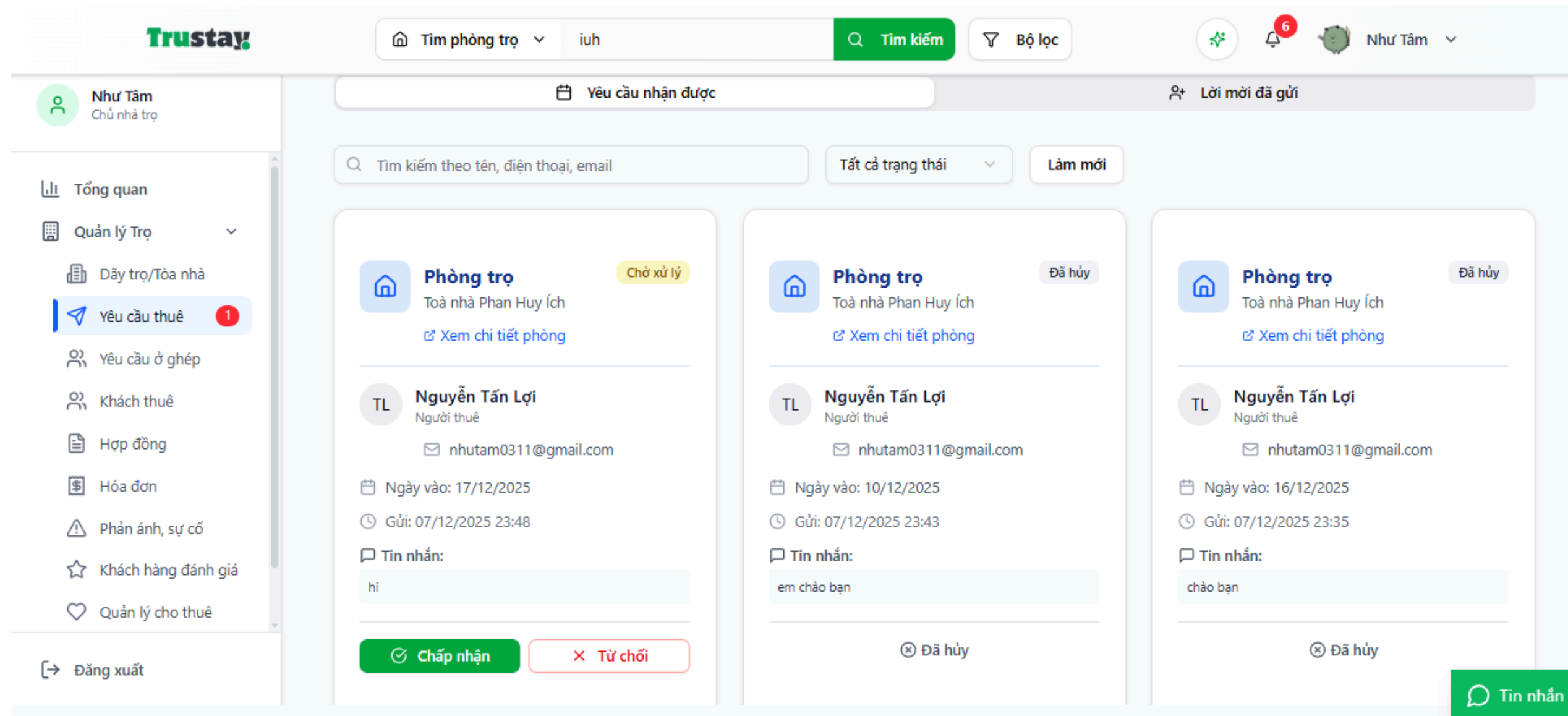


Dashboard Landlord screen

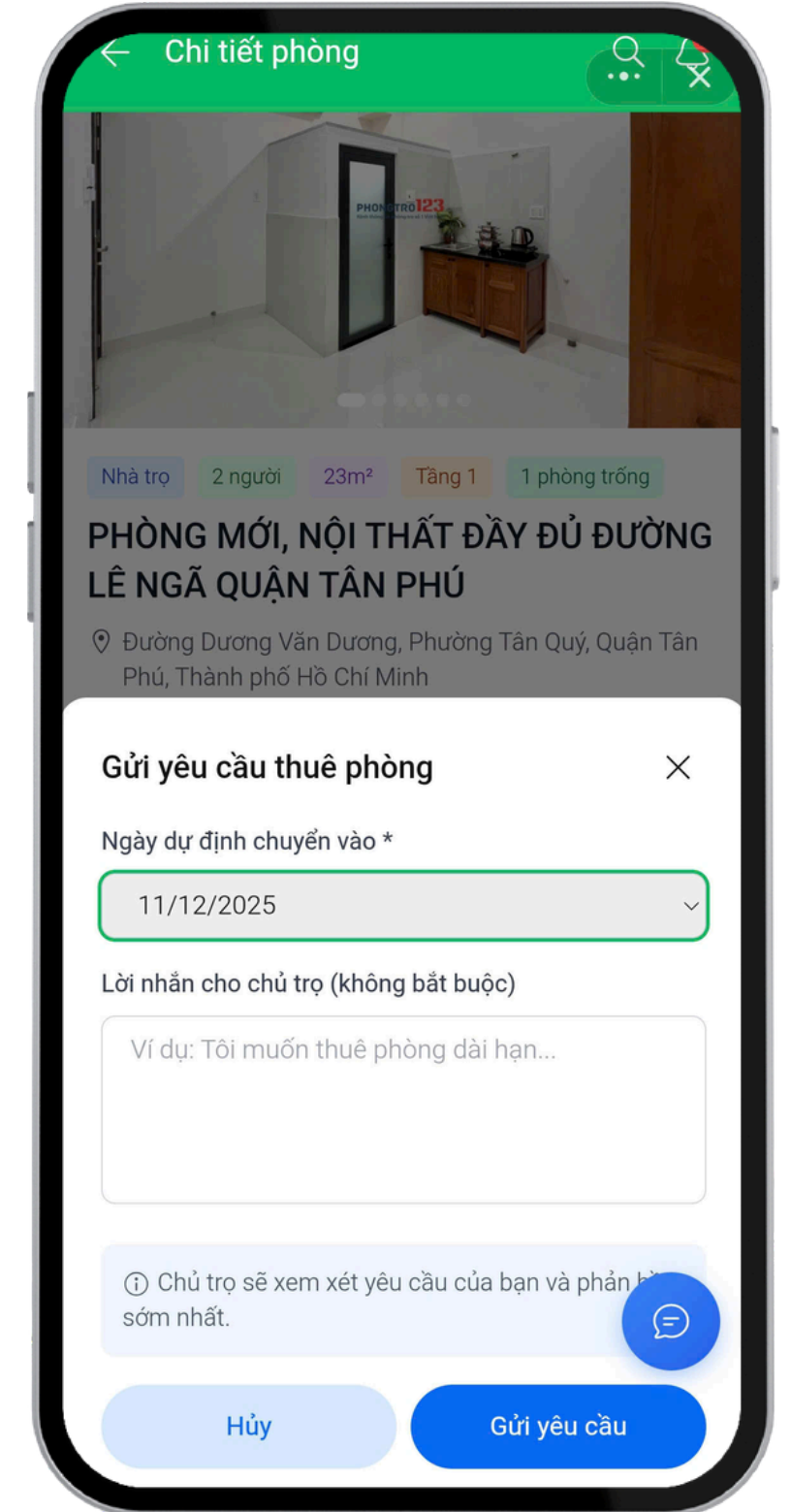


Dashboard Tenant screen

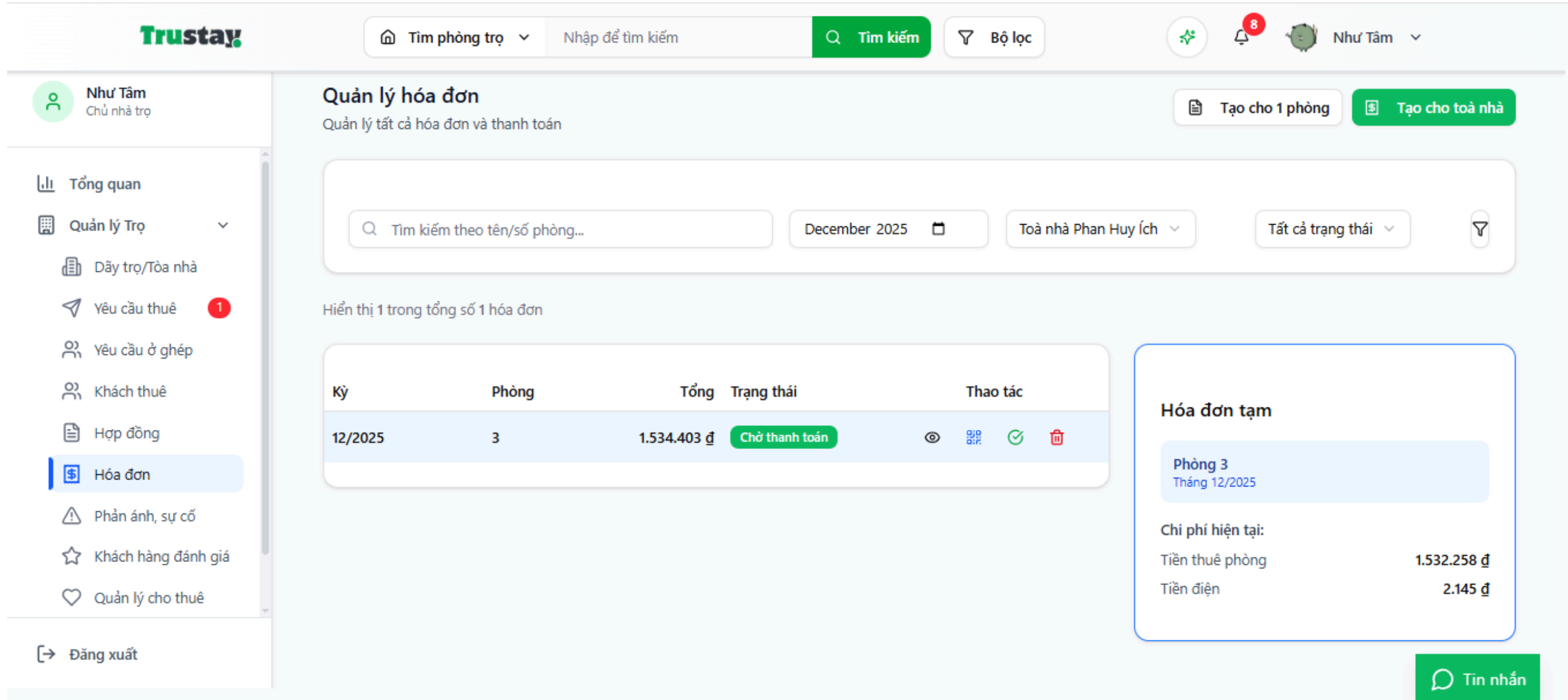




Room bookings screen







**Trustay**

Tìm phòng trọ | Nhập để tìm kiếm | Tìm kiếm | Bộ lọc

Như Tâm | Chủ nhà trọ

**Quản lý hóa đơn**  
Quản lý tất cả hóa đơn và thanh toán

Tạo cho 1 phòng | Tạo cho toà nhà

Tìm kiếm theo tên/số phòng... | December 2025 | Toà nhà Phan Huy Ích | Tất cả trạng thái

Hiển thị 1 trong tổng số 1 hóa đơn

| Kỳ      | Phòng | Tổng        | Trạng thái     | Thao tác  |
|---------|-------|-------------|----------------|-----------|
| 12/2025 | 3     | 1.534.403 đ | Chờ thanh toán | 👁️ 📄 ✅ 🗑️ |

**Hóa đơn tạm**

Phòng 3  
Tháng 12/2025

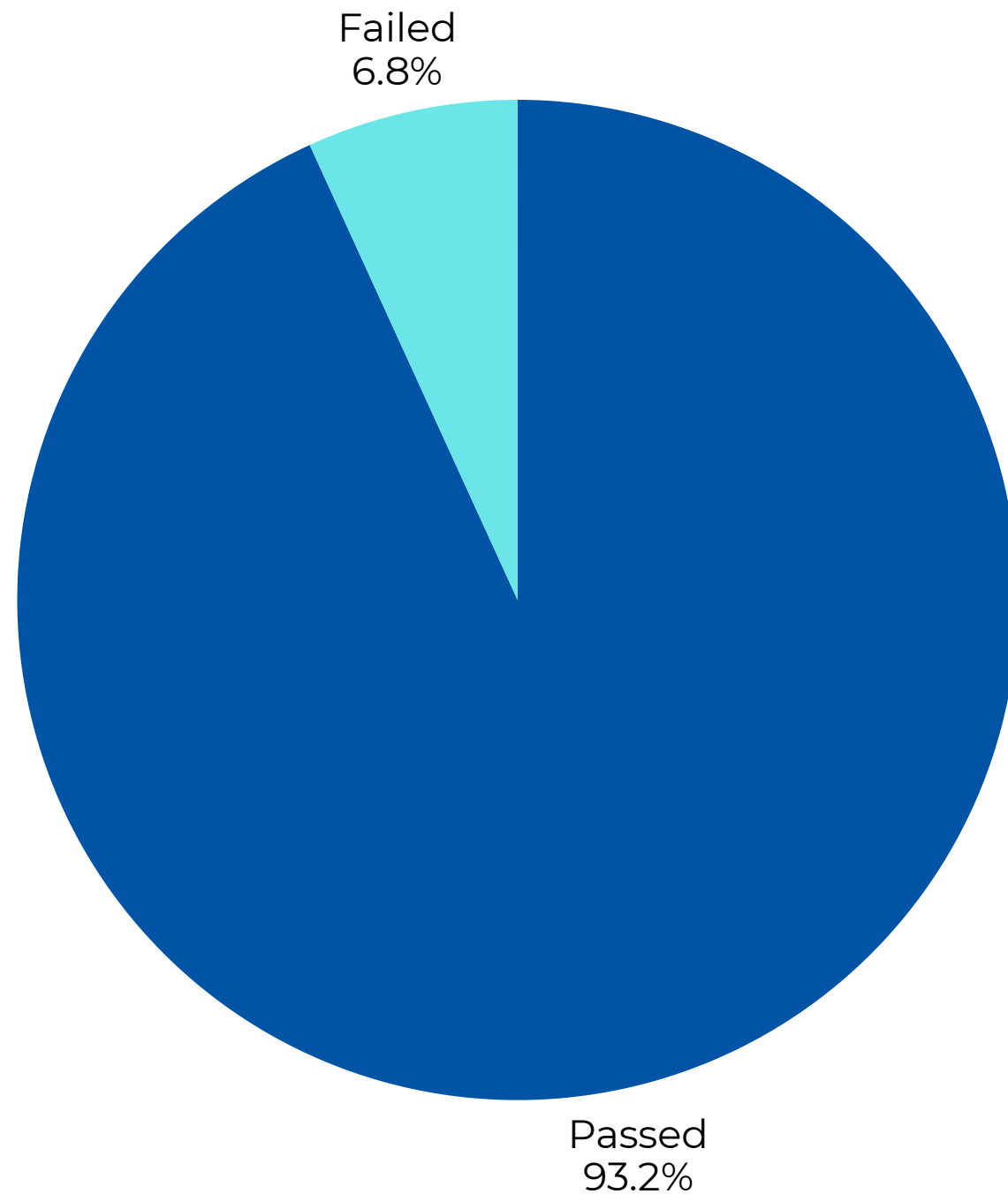
Chi phí hiện tại:

|                 |             |
|-----------------|-------------|
| Tiền thuê phòng | 1.532.258 đ |
| Tiền điện       | 2.145 đ     |

Tin nhắn

Payment and invoice management screen



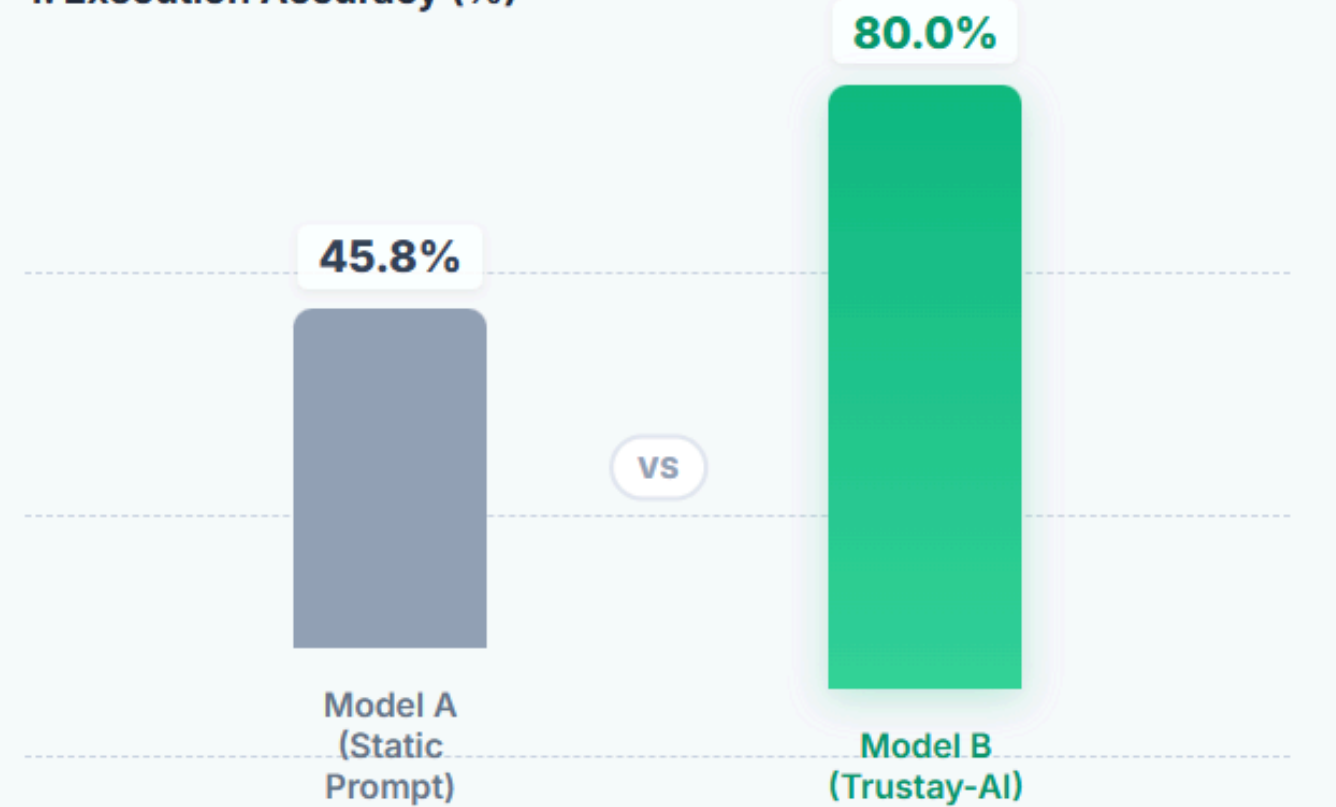


### Testing Result:

The system operates stably across core functions: Contracts, Payments, and AI Chat.



### 1. Execution Accuracy (%)



### 2. Canonical SQL Reuse Efficiency

Metric: Avg Latency

Comparison: No Reuse & Reuse Canonical SQL



**~33.4% Faster Response**  
for repetitive queries via Reuse Canonical SQL



#### Significant Gain (+34.2%)

Trustay-AI outperforms the baseline by handling context & self-correction effectively.



#### Canonical Reuse

Valid SQLs are stored as "Golden Data," reducing latency for recurring intents.



#### Zero-to-Golden

System automatically harvests "Golden Data" during operation, solving the Cold Start problem.



### Achievements

- Successfully built a complete Web platform and Zalo Mini App
- Successfully applied AI (RAG + Multi-Agent) for natural-language data querying
- Digitized the entire rental workflow (Contracts, Invoices, Payments)
- System deployed stably on the cloud with CI/CD automation



### Limitations

- Multi-Agent model accuracy still needs improvement
- Legal digital signature (eKYC) has not been integrated



## 05. Result & Future work



Provide optimal rental price recommendations based on market data.



Integrate additional e-wallets (Momo, ZaloPay). Add ZNS notifications.



Integrate eKYC (National ID verification) to enhance the legal validity of contracts.



Industrial  
University of  
Ho Chi Minh City

**Thank You for Your Attention!**